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Expt.No: 4

Title: Python program to perform Basic operations on an Image

In [1]:

```
import cv2
import matplotlib.pyplot as plt

image = cv2.imread('ResizeImage3.jpg')

x, y, w, h = 100, 100, 300, 300
cropped_image = image[y:y+h, x:x+w]

scaled_image = cv2.resize(image, (800, 800))

rotated_image = cv2.rotate(image, cv2.ROTATE_90_CLOCKWISE)

flipped_image_horizontal = cv2.flip(image, 1)
flipped_image_vertical = cv2.flip(image, 0)

plt.figure(figsize=(12, 8))

plt.subplot(2, 3, 1)
plt.imshow(cv2.cvtColor(image, cv2.COLOR_BGR2RGB))
plt.title('Original Image')
plt.axis('off')

plt.subplot(2, 3, 2)
plt.imshow(cv2.cvtColor(cropped_image, cv2.COLOR_BGR2RGB))
plt.title('Cropped Image')
plt.axis('off')

plt.subplot(2, 3, 3)
plt.imshow(cv2.cvtColor(scaled_image, cv2.COLOR_BGR2RGB))
plt.title('Scaled Image')
plt.axis('off')

plt.subplot(2, 3, 4)
plt.imshow(cv2.cvtColor(rotated_image, cv2.COLOR_BGR2RGB))
plt.title('Rotated Image')
plt.axis('off')

plt.subplot(2, 3, 5)
plt.imshow(cv2.cvtColor(flipped_image_horizontal, cv2.COLOR_BGR2RGB))
plt.title('Flipped Image (Horizontal)')
plt.axis('off')

plt.subplot(2, 3, 6)
plt.imshow(cv2.cvtColor(flipped_image_vertical, cv2.COLOR_BGR2RGB))
plt.title('Flipped Image (Vertical)')
plt.axis('off')
plt.show()
```


Original Image
Cropped Image
Scaled Image
Rotated Image
Flipped Image (Horizontal)
Flipped Image (Vertical)

In [ ]:

```
```